Properties of logarithms

1

a $\log 36$ **b** 2 **c** $\log_{10} 198$ **d** $\log_{10} 2$

e 2 **f** 3 **g** 2 **h** 9

i $2 + \log xy$ **j** $5 \log x$ **k** -4 **l** 5

m 5 **n** 2 **o** w **p** $\dfrac{81}{64}$

2

a 2 **b** $\dfrac{48}{5}$ **c** 12 **d** $\dfrac{\ln x}{\ln (y^5)}$

3

a $\log_9 u + \log_9 v$ **b** $4 \log x$

c $7 \log_4 x$ **d** $\log_5 9 - \log_5 7$

e $\dfrac{2}{5} \log x$ **f** $2 \log_b x$

g $\log 3 - \log x$ **h** $5 (\log 3 + \log x)$

i $\log 7 - 4 \log x$ **j** $-7 \log 5 - 7 \log x$

k $-(\log 2 + \log x)$ **l** $-\log x - \log y$

m $\log p + \log q - \log r$ **n** $-\log (3x + 7)$

o $-8 \log (3x + 4)$ **p** $\dfrac{1}{2} \log (5x + 7)$

q $6 \log (5x + 2)$ **r** $5 \log (x + 6)$

s $\log_2 5 + \log_2 x$ **t** $\log_{10} 2 - \log_{10} x$

u $\log m + \log m$ **v** $\log 5 + \dfrac{2}{3} \log x$

w $\dfrac{1}{3} (\log 14 + \log x)$ **x** $4 \log c - \dfrac{1}{2} \log d$

4

a $\ln 15 - \ln 8$

b $\ln p + \ln q + \ln \left(\dfrac{1}{r}\right)$ or $\ln p + \ln q - \ln r$

c $\ln 19 - \ln 7$

5

a $7 \log x$ **b** $\log \left(x^5 y^3\right)$ **c** $\log \left(\dfrac{x^8}{y^{\frac{1}{3}}}\right)$ **d** $\log \left(\dfrac{x^8}{y}\right)$

e $-35 \log_{10} 5$ **f** $9 \log_{10} 2$ **g** 2 **h** 2

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

6

a $\ln 15$ **b** $\ln 6$ **c** $\ln\left(\frac{5}{7}\right)$ **d** $\ln 440$

e $\ln 2$ **f** $\ln\left(\frac{42}{5}\right)$

7

a $\ln\left(\frac{x^6}{y^{\frac{1}{6}}}\right)$ **b** $\ln(15xy)$ **c** $5\ln x$ **d** $7\ln x$

8

a $2\ln y$ **b** $-2\ln y$ **c** $\frac{1}{2}\ln y$ **d** $\frac{1}{3}\ln y$

e $-5\ln y$ **f** $\frac{2}{9}\ln y$ **g** $\frac{1}{2}\ln y$ **h** $\frac{1}{2}\ln w$

9

a $\ln\left(\frac{1}{x^2}\right)$ **b** $\ln\left(x^{\frac{1}{2}}\right)$

Change of base

10

a $\frac{\log_{10} 16}{\log_{10} 4}$ **b** $\frac{\log_{10} 0.9}{\log_{10} 3}$ **c** $\frac{\log_{10} \sqrt{5}}{\log_{10} 3}$ **d** $\frac{\log_{10} B}{\log_{10} a}$

11 $\frac{\log_4 20}{\log_4 3}$

12

a **i** $\frac{\log_{10} 21}{\log_{10} 8}$ **ii** 1.46

b **i** $\frac{\log_{10} \sqrt{5}}{\log_{10} 4}$ **ii** 0.58

c **i** $\frac{\log_{10} 3^3}{\log_{10} 2}$ **ii** 4.75

d **i** $\frac{\log_{10} 105}{\log_{10} \pi}$ **ii** 4.07

© ClipSAT: By Mr. Mohamed Abdallah +966597688647